

PARTICULAR SPECIFICATION P7

Building Services

CONTENTS

P7.1	Introduction	1
P7.1.1	Overview	1
P7.2	Air-Conditioning and Mechanical Ventilation (ACMV)	1
P7.2.1	General.....	1
P7.2.2	Buildings / Rooms Mechanical Ventilation (MV)	2
P7.2.3	Buildings/Rooms to be Provided with Air-Conditioning (AC)	3
P7.3	Automation Lubrication System (ALS)	4
P7.3.1	Performance Requirements of ALS.....	4
P7.4	Building Management System (BMS).....	5
P7.4.1	BMS Scope and Overall Requirements	5
P7.4.2	BMS Functional Requirements.....	8
P7.4.3	BMS System Architecture	9
P7.4.4	BMS Software Engineering.....	10
P7.4.5	BMS Cabling	11
P7.4.6	Local Control Station.....	11
P7.5	Integrated Security Management System (Access Control and Security CCTV)	12
P7.6	Fire Protection, Detection and Alarm System.....	13
P7.6.1	General.....	13
P7.6.2	Fire Alarm System	15
P7.6.3	Gas Suppression System	15
P7.6.4	Fire Hose Reel System	17
P7.6.5	Dry Riser System	17
P7.6.6	Public Address System	17
P7.6.7	Fire Extinguishers	18
P7.7	Plumbing and Sanitary Systems	18
P7.7.1	General.....	18
P7.7.2	Sanitary System.....	18
P7.7.3	Water Supply Plumbing System.....	19
P7.8	Testing and Commissioning	19

P7.1 INTRODUCTION

P7.1.1 Overview

- (a) Building services work shall be provided in the following locations:
 - (i) Digestion Pre-Treatment (WSD11);
 - (ii) Sidestream Nitrogen Removal (WST02).
- (b) Building services shall comply with General Specification G7 Building Services. Reference shall be made to all other Particular and General Specifications.
- (c) The Drawings and Specifications describe notional design for building services and the Specifications provide minimum performance requirements.
- (d) Contractor's design and build scope shall include, but not limited to:
 - (i) Detailed Calculations;
 - (ii) Pipe hydraulic calculations;
 - (iii) Selection of equipment and materials;
 - (iv) Verification of final pressure and voltage drop for services;
 - (v) Verification of effective water storage facilities;
 - (vi) Acoustic, vibration, support, plinth and bracket design for equipment and pipes, ducts, cables and etc.;
 - (vii) Detailed final layout drawings of sprinkler, hose reels and hydraulic calculations of such; and
 - (viii) All the above to be submitted to the S.O. for final approval.
- (e) The Contractor shall engage the appropriate parties and Fire Engineer (FE) for obtaining all permits in relation to engineered smoke control system including but not limited to TFP, FSC, TOP and CSC for the works.

P7.2 AIR-CONDITIONING AND MECHANICAL VENTILATION (ACMV)

P7.2.1 General

- (a) This section specifies the requirements necessary for the provision of fully functional Air-Conditioning and Mechanical Ventilation (ACMV) systems.
- (b) The ACMV System shall be monitored and controlled by the Building Management System (BMS) for normal operation and fire mode operation.

- (c) The ACMV shall comply with the latest energy efficiency standard for building services spelt out under SS530.
- (d) Access platforms, ramps and steps over pipes and trunkings on floor and ladders shall be provided to facilitate necessary passage access to plant locations requiring to be accessed and to safely access all items of plant including equipment, valves and instruments. The platforms, stairways and ladders shall be designed taking into consideration any special access requirement that may be associated with all specific items of plant in question with respect to repair, maintenance, cleaning, greasing/lubricating, draining, replacement, adjustment, dismantling, hoisting operations, assembly, etc. The configuration of the platform, stairway and ladder shall be designed to suit each plant item and location requiring them.
- (e) The Contractor shall appoint a Professional Engineer to carry out the design of all relevant support elements and shall obtain approval from relevant statutory or Government authorities, such as BCA, after liaising with S.O. and Accredited Checker. The cost of design of support system is deemed to have been included in the Contract sum.

P7.2.2 Buildings / Rooms Mechanical Ventilation (MV)

- (a) Mechanical ventilation shall be provided as per Fire Code requirements.
- (b) All non-naturally ventilated escape staircases, if required, shall be provided with MV with emergency power supply to achieve a minimum of four (4) air changes per hour (ACH) under normal ventilation to SS 553 latest edition, and fire mode staircase pressurisation shall maintain a pressure differential of not less than 50 Pa between the pressurised exit staircase and the occupied area when all doors are closed to SCDF Code of Practice for Fire Precautions in Buildings. The MV fans shall be located on the Roof Level and shall be linked to the fire alarm system and Fire Command Centre (FCC) at WIL11 (Contract 3A) for activation of staircase pressurisation during fire mode operations. Pressure relief damper(s) are to be provided in each staircase to regulate the pressure as per Fire Code. Sufficient airflow at required velocities through door openings as per Fire Code is to be provided.
- (c) Smoke stop lobby, if required, MV system with emergency power supply shall be provided to achieve a minimum of four (4) ACH under normal operation and a minimum of ten (10) ACH under fire mode operations. The MV fans shall be linked to the fire alarm system and FCC for activation of fire mode operation.
- (d) Redundancy as per Fire Code is to be provided.
- (e) All motorised dampers shall be provided with an external red indicator to indicate when the damper is in its open or close position. The indicator shall align with the ductwork when in the open position.
- (f) MV systems with redundancy fans shall be programmed in the BMS on an equal duty (run time) basis for rotation of duty/standby operation.

- (g) All ductwork shall be SS304 sheet metal ductwork with flanged type connection including recommended and approved gaskets and sample to be submitted for S.O.'s approval.
- (h) Battery Rooms:
 - (i) MV systems (exhaust only) for Battery Rooms are to be connected with secondary power supplies, MV systems are to provide a minimum of six (6) ACH and shall be activated by a hydrogen sensor in the event the hydrogen gas level exceeds 1% of the total volume of the Battery Room; the MV system shall be connected to the Fire Alarm Panel and display panel (with indication "exhaust fan activated due to high hydrogen level" with alarm) at entrance of the room as per Fire Code.
 - (ii) A 'fail-safe' natural ventilation system shall be provided for Battery Rooms as a secondary source of room ventilation in the event of a power failure. Motorised dampers shall 'fail-open' during power failure to ventilate the room and prevent hydrogen accumulation while power mains are down as per Fire Code.

P7.2.3 Buildings/Rooms to be Provided with Air-Conditioning (AC)

- (a) AC shall be provided, as minimum, at these areas:
 - (i) Switchrooms;
 - (ii) Battery Rooms;
 - (iii) UPS Rooms;
 - (iv) Process XLV Rooms;
 - (v) Process non-XLV Rooms; and
 - (vi) Operations Room.
- (b) AC shall be designed to achieve the following performance inside buildings/rooms. Indoor air-conditioning design condition to be:
 - (i) Design Dry Bulb: 24 °C to 26°C;
 - (ii) Relative Humidity: <65%; and
 - (iii) Fresh air supply to maintain the desired indoor air quality in accordance with SS553 and SS554. For total cooling to the space/rooms more than 35kW, the fresh air supplied to the air-conditioned space shall be filtered, cooled and dehumidified.
- (c) AC shall be provided by direct expansion (DX) air-cooled type consisting units serving both the Variable Refrigerant Flow (VRF) type AC equipment (Indoor AC Units / Outdoor Units) and packaged DX Air Handling Units (AHU) evaporator coils.

- (d) 100% redundancy shall be provided for AC VRF systems for Switchrooms, Battery Room, UPS Room, Process XLV room and Non-Process XLV room.
- (e) The following outdoor ambient conditions are to be assumed in the design of air conditioning systems:
 - (i) Design Dry Bulb: 33 °C; and
 - (ii) Wet Bulb Temp: 28 °C
- (f) Condensate drain pipes provided from AC equipment shall be connected to the floor traps as per S.O.'s approval, complete with insulation. Floor traps with condensate drain connection are to be insulated and the waste pipe to be insulated to a minimum of 10m length.
- (g) All SS304 sheet metal ductwork installed shall be of flanged type complete with external minimum 25 mm thick rockwool insulation and double-sided vapour barrier.
- (h) All equipment shall have adequate space and hoisting facilities for maintenance and replacement of fans etc.
- (i) Final equipment and locations shall be selected with due regards for safe and efficient access and serviceability and confirmed with S.O..
- (j) Access platforms and walkways shall be designed in accordance with BS EN ISO 14122-2. The minimum design loads shall be:
 - (i) 2kN/m² uniformly distributed; and
 - (ii) 1.5kN concentrated load applied in the most unfavourable position over an area of 200 mm x 200 mm of the floor.
- (k) Personal fall arrest systems shall be provided for platform and roof maintenance access.

P7.3 AUTOMATION LUBRICATION SYSTEM (ALS)

P7.3.1 Performance Requirements of ALS

- (a) Bearing lubrication requirements of all fans, blowers, pumps and other equipment requiring lubrication shall be fulfilled by ALS specified herein.
- (b) The Contractor shall provide local ALS to automatically lubricate the bearings of the fans, blowers, pumps, motors and motor-to-pump shaft sections that require regular lubrication for all ACMV and fire protection equipment.

- (c) Each ALS shall consist of replaceable or rechargeable oil or grease dispenser units/canisters and where necessary, extension tubes. The requirements for the lubrication system shall include, but not be limited to, the following:
 - (i) Lubricants (oil or grease) and lubrication frequencies shall comply with machinery and equipment manufacturer's requirements. Grease lubricating systems shall be designed to permit safe lubrication of the machinery while it is operating;
 - (ii) Oil or grease dispenser units shall give a clear indication of grease content. The oil or grease dispenser unit capacity shall be selected to provide the required degree of lubrication for a minimum of twelve (12) months;
 - (iii) Extension tubes shall be fitted to the lubrication points to allow convenient positioning of, and access to, grease dispenser units, for replacement/recharging; and
 - (iv) The initial priming of any extension tubes shall be undertaken prior to fitting of the grease dispenser units.
- (d) The Contractor with the aid of the ALS supplier and equipment/pump suppliers shall select from the available types and configurations of ALS to cater to the lubrication requirements of the equipment and pumps.

P7.4 BUILDING MANAGEMENT SYSTEM (BMS)

P7.4.1 BMS Scope and Overall Requirements

- (a) The BMS shall be implemented using industrial-grade PLCs and using a BMS SCADA system.
- (b) ABB BMS SCADA system will be provided by Contract 5A.
- (c) The BMS PLC and BMS SCADA shall comply with General Specification G7 Building Services. The BMS shall comply with all applicable local codes, regulations, standards and requirements issued by all the local statutory authorities, and services providers.
- (d) The Contractor shall provide a complete and fully functional BMS complying with General Specification G7 and Drawings, and comprising :
 - (i) Main BMS Network switches (BNS);
 - (ii) BMS Edge network switches (BES);
 - (iii) BMS Programmable logic controllers (PLC), having conformally coated circuit boards, also referred to as "BPLC";

- (iv) BMS I/O (BMS RIO), having conformally coated circuit boards;
 - (v) Laptop computer complete with BMS PLC programming software;
 - (vi) All equipment shall be connected to the UPS supply.
 - (vii) An additional BMS PLC programming software license for installation on the BMS SCADA server;
 - (viii) FIBOTs, fibre and copper cabling; and
 - (ix) Panels and racks to house the above equipment.
- (e) The Contractor shall utilise BMS Supervisory Control and Data Acquisition (SCADA) Server and workstation hardware and software licenses of the BMS SCADA platform. The BMS SCADA Platform is ABB, provided by Contract 5A. It will be an industrial-grade SCADA system, scalable to at least 100,000 tag points. The Contractor shall be responsible for the installation of the software on the servers and workstations, however the software licenses will be provided by Contract 5A.
- (f) The Contractor shall be responsible for the configuration of BMS SCADA database and communications settings.
- (g) The Contractor shall provide software engineering for BMS PLC programming and BMS SCADA configuration and for BNS and BES switch configurations, including necessary software changes.
- (h) BMS SCADA templates will be provided by the S.O. after Award. These standard templates shall be used for the BMS SCADA configuration; however, the Contractor shall develop any additional templates found to be required.
- (i) The BMS shall receive, react to, and return information to multiple building systems, including but not limited to the chillers, boilers, pumps, fans, AHU's, variable frequency drives, power monitoring system.
- (j) Inputs and outputs from the building systems shall be configured in the BMS system to provide the appropriate real-time interoperability including automatic and time scheduled control, energy management, alarms, trends and historical data analysis, totalization.
- (k) The automatic control of building systems shall be independent of the status of the BMS SCADA. Systems that rely on an operator workstation or servers to provide execution of the sequences of operations shall not be acceptable.
- (l) BMS Graphic Displays
- (i) The Contractor shall configure a colour graphic system flow diagram display for each building system with all points as indicated on the point list. All terminal unit graphic displays shall be from a standard design library provided by the Employer.

- (ii) Users shall be able to access the various system schematics via a graphical overview scheme and by a selection menu selection.
- (m) Progressive, interim, and final versions of BMS PLC code and BMS SCADA configurations shall be provided to the Board via the S.O., in electronic source code format. These shall include libraries of PLC function blocks and SCADA faceplates and graphical object templates, and SCADA databases. The S.O. will subsequently provide this material in native format to other Contractors to ensure consistency across the whole of the TWRP project.
- (n) The Contractor shall interface the BMS SCADA with Fire Alarm Panels.
- (o) The Contractor shall interface the BMS with the site-wide Non-Process Network Backbone via the network switch provided by Contract 5A.
- (p) The Contractor shall liaise with Contract 2A to integrate the BMS systems provided with the IPS BMS System:
- (q) The integration shall achieve a single integrated BMS system retaining the original facility-specific servers. The original facility-specific workstations by Contract 2A shall all become shared common workstations, each able to access BMS functions for any part of the plant.
- (r) BMS shall comply with the standards as per Table P7.3 below:

Table P7.3-1 : Standards

Description	Standards
Environmental Testing	IEC 60068
Electrical Relays	IEC 60255
Programmable Logic Controllers	IEC 61131-1 IEC 61131-2 IEC 61131-3 IEC 61131-4 IEC 61131-5 IEC 61131-8
Electromagnetic Compatibility (EMC)	IEC 6100

- (s) The Contractor shall provide one BMS Hot Standby PLC as specified in the Drawings.
- (t) The Contractor shall provide adequate BMS I/O to function as described. I/O for duty and standby equipment shall be divided across different I/O modules.

- (u) BMS PLC I/O is not required to have HART functionality.
- (v) The network shall provide high speed (100 Mb/s) communication using Ethernet TCP/IP.
- (w) All damper and valve shall be electrically actuated.
- (x) All air handlers shall be controlled by the BMS PLC.
- (y) Terminal Units (VAV, UV, etc.) shall have electric damper and valve actuation; and shall be controlled by the BMS PLC.
- (z) The system shall comply with the Board's General Cybersecurity Requirements.

P7.4.2 BMS Functional Requirements

- (a) The BMS system shall monitor and control the building services equipment including but not limited to:
 - (i) Monitor Only
 - 1) Fire Alarm System – For each alarm panel: Ready, Fault, Alarm activated, Isolated zones status, individual detector and break glass status.
 - (ii) Control and Monitor
 - 1) Air- Cooled Packaged / AHU's –individual AHU running, stopped, speed & fault, available, power on, local mode selected, BMS mode selected, individual AHU control, DX coil power on, off-coil temperature, relative humidity;
 - 2) MV Fans – individual MV Fans running, speed & fault, available, power on, local mode selected, BMS mode selected, individual fan control; and
 - 3) FCU – individual FCU running, stopped, fault, individual FCU control and setpoint.
- (b) The BMS system shall have the necessary inputs and sensors to allow long-term tracking by the Power Monitoring System (PMS) of the specific energy consumption of individual equipment items. e.g. kW·hr/kl pumped, kW·hr/m³ air, kW/RT cooling. The PMS itself will be provided by Contract 5A, however the BMS shall provide all data points such as flows, pressures, temperatures, humidity, equipment running status, to the PMS via the Non-Process Backbone.

P7.4.3 BMS System Architecture

- (a) The BMS network architecture, consisting of BNS and BES switches, shall support star topology and shall be fault-tolerant. All BMS communications shall use ethernet networks, not serial communications.
- (b) The BMS network shall connect to the Non-Process Backbone Switches provided by Contract 5A.
- (c) No single I/O panel shall control or monitor more than half the equipment items of each type
- (d) The BMS shall have graphical schematic representations of the building services being controlled and monitored, with trend, alarm and event logging pages, and pages to show the status of various BMS components (e.g. servers, switches, etc). The graphics shall be arranged to facilitate the safe and efficient operation of the building services equipment. BMS SCADA software shall provide the elements necessary to safely and efficiently operate the building services equipment. The BMS shall be configured for multiple levels of user-defined security, user-defined span-of-control definition, fully-functional graphics, alarm and event management, real-time trending.
- (e) The BMS PLC shall also comply with the following:
 - (i) PLC make, and model shall be from an internationally reputable manufacturer with a wide installation base in Singapore, including in municipal water treatment installations in Singapore, and shall have a proven track record;
 - (ii) Be a microprocessor-based programmable device with chassis-mounted plug-in modules, and shall comprise of CPU processors, power supply modules, communication modules, input / output modules, third party interface modules, logic memory, data storage memory and shall have the necessary data communication facilities to provide correct interfacing with the building services equipment to achieve the specified system operation;
 - (iii) Have a high availability, hot-standby architecture. There shall be hot-standby processors, power supply modules, and communications modules, with duty and standby modules being split between two chassis. No failure of any single component shall cause any loss of function except that failure of input/output (I/O) module may cause loss of functions associated with that I/O module;
 - (iv) Have hot-swappable input and output modules;
 - (v) Hot-standby arrangements shall natively achieve automatic bump-less transfer in the event of the failure of any primary unit;

- (vi) The redundancy features shall be a hardware built-in capability such that there is no need for customised programming of the redundancy capability;
 - (vii) PLC shall be capable of connection and incorporation into a control network together with remote IO modules;
 - (viii) Have a minimum of 20% spare programme memory and 20% spare data register memory shall be available when final commissioning is completed. In addition, the memory shall be expandable, using expansion modules, to provide a further increase in capacity of at least 50%;
 - (ix) Have 20% spare channels of each IO type; and
 - (x) PLCs and RIO shall be provided at locations to facilitate efficient interfacing with the field equipment. The locations of the RIO panels shall be subject to acceptance by the S.O..
- (f) BMS PLC shall be provided with diagnostic capabilities to identify failures to the module level. Diagnostic indications at BMS SCADA shall include but not be limited to notification of processor failure, input or output module failure, communications failure, and battery low.
- (g) Integration with mechanical plant shall be by the following methods:
- (i) Hardwired analogue and digital hardwired signals to BMS PLC i/o; and
 - (ii) Bi-directional data communications using one of the following protocols:
 - 1) Modbus TCP/IP;
 - 2) BacNet IP;
 - 3) Ethernet/IP;
 - 4) Profinet; and
 - 5) OPC.

P7.4.4 BMS Software Engineering

- (a) The Contractor's BMS PLC programming shall comply with the following:
- (i) BMS PLC's shall be programmed using function block diagram (FBD) and sequential function chart (SFC) languages;
 - (ii) BMS PLC software structure and organisation shall adhere to an object-oriented methodology whereby a control code for individual units must be identical in a multiple unit process system. A control logic modification shall be done once and populated to all units;

- (iii) Code documentation shall include an overview description and control objective within each section of code. The sections, subsections, and function block of the programs shall be provided with descriptions to help other to understand the programs;
 - (iv) Variable tag names related to plant asset tags shall be used instead of memory addresses;
 - (v) A library of user defined function blocks shall be created for common equipment (such as pumps, valves, control loops sequences, etc.) and systems. The functionality of these function blocks shall comply with that specified in the Software Standards Specification which will be issued after Award. These standard blocks shall be used for all BMS PLC programming, and shall be provided to the S.O., complete with software testing documentation to the satisfaction of the S.O., for use by Other contractors; and
 - (vi) A library of user defined BMS SCADA faceplates and graphical object templates shall be provided for common equipment and systems. These shall comply with the Software Standards Specification which will be issued after Award. These standard blocks shall be used for all BMS SCADA configuration, and shall be provided to the S.O., complete with software testing documentation to the satisfaction of the S.O., for use by Other contractors.
- (b) The Contractors BMS SCADA configuration shall be in accordance with the requirements of the SCADA platform nominated by the S.O..
 - (c) PLC and SCADA software developed and used shall become the property of the Board. The programs shall not have security that prevents the Board from modifying the programs.

P7.4.5 BMS Cabling

- (a) Fibre optic cabling shall be provided as required to achieve the BMS depicted in the Drawings. Refer to General Specification G6.2 Installation Requirements for cabling requirements.
- (b) Cat 6A FTP cable shall be provided for communications between PLC to field devices complete with cable support, conduits, trunking and associated accessories. Refer to General Specification G6.2 Installation Requirements for cabling requirements.
- (c) Other cabling shall be provided as required to form a fully functioning and BMS.

P7.4.6 Local Control Station

- (a) Local control station (LCS) shall be provided as specified in the Drawings. For inaccessible building services equipment, LCS shall be remote type and shall be placed in accessible locations within operators' reach.

P7.5 INTEGRATED SECURITY MANAGEMENT SYSTEM (ACCESS CONTROL AND SECURITY CCTV)

- (a) The Contractor shall provide ISMS cabling for all buildings complete with trunking, cable trays, conduits, junction and knock-out boxes for the subsequent connection and installation by Other contractors of the access control system, including the biometric access readers, access controller and access electromagnetic lock. The access reader cables shall be provided up to the network switch locations in the relevant Non-process XLV rooms (as shown in the security CCTV schematics). Refer to General Specification G7 Building Services for the requirements for cabling.
- (b) The Contractor shall provide cabling as required for access control system interfaces between buildings. Refer to General Specification G6.2 Installation Requirements for ELV and fibre optic cabling requirements.
- (c) These areas, as minimum, shall be provided with access control as per the concept described below. The Contractor shall submit the access control plan and final plan is subjected to S.O.'s approval.
 - (i) At WST02 Sidestream Nitrogen Removal Facility, all single-leaf and double-leaf door at ground floor that provide access from external to interior of the building shall be provided with biometric reader complete with magnetic lock and break glass at interior;
 - (ii) At WST02 Sidestream Nitrogen Removal Facility, roller shutters shall be provided with in-vehicle unit;
 - (iii) At WST02 Sidestream Nitrogen Removal Facility, the gates at all transformer bays shall be locked by key;
 - (iv) At WST02 Sidestream Nitrogen Removal Facility, all switch rooms shall be provided with one door that is access with biometric reader complete with magnetic lock and break glass at interior and the rest of the doors to be fire escape door;
 - (v) At WST02 Sidestream Nitrogen Removal Facility, all fire escape door shall be provided with fire escape bar and key; and
 - (vi) At WST02 Sidestream Nitrogen Removal Facility, all battery rooms, UPS rooms, XLV process rooms, XLV non-process rooms, cylinder gas rooms shall be provided with biometric reader complete with magnetic lock and break glass at interior.
- (d) The Contractor shall provide a CCTV cabling system for security and plant CCTV complete with trunking, cable trays, conduits and junction boxes for the connection between the CCTV IP (PoE) camera locations and the Non-Process XLV Room. Refer to General Specification G7 Building Services for the requirements for cabling.

- (e) The CCTV will be provided by Other contractors. CCTV shall be located, as minimum, at these areas as per the concept described below. The Contractor shall submit the CCTV plan and final CCTV location is subject to coordination with Other contractors and S.O.'s approval:
 - (i) At WST02 Sidestream Nitrogen Removal Facility, the security cameras shall be placed at all indoor location to monitor all ingress and egress into the building;
 - (ii) At WST02 Sidestream Nitrogen Removal Facility, the security cameras shall be placed within these rooms to monitor all ingress and egress into the rooms:
 - 1) Battery rooms;
 - 2) UPS rooms;
 - 3) XLV process rooms; and
 - 4) XLV non-process rooms.
 - (iii) At WST02 Sidestream Nitrogen Removal Facility and WSD11 Digestion Pre-treatment, the plant CCTV cameras shall be placed at indoor or outdoor locations to monitor all major process mechanical equipment and building services equipment provided under this Contract.
- (f) Fibre optic (FO) cabling shall be tested and ready for termination by Other contractors. Refer to Particular Specification P6 ICA Works for fibre optic cabling testing & commissioning requirements. Refer to General Specification G6 Installation Requirements for fibre optic cabling requirements.
- (g) In general, FO cables from edge switch/PoE Network Switches/patch panel to fibre patch panel shall be minimum 12 core single mode and from fibre patch panel to backbone switch in same XLV room shall be minimum of 2x6 core multi-mode cable.
- (h) Provide Cat 6A FTP cabling from CCTV camera locations to PoE Network Switches/patch panels at XLV rooms. At camera locations, allow for minimum 10m slack coil for final run and termination of cameras by Other contractors.
- (i) Provide floor standing 19" rack enclosures and patch panels in Non-Process XLV rooms for termination of Cat 6A FTP and fibre optic cabling.

P7.6 FIRE PROTECTION, DETECTION AND ALARM SYSTEM

P7.6.1 General

- (a) The Contractor shall provide a fully functional fire detection and protection system to meet the requirements of all relevant Singapore Standards (SS), Codes of Practice (CP), Building Control Regulations, Fire Safety Regulations, Fire Code and other requirements of the relevant authorities.

- (b) The Contractor shall also refer to General Specification G7.4 Fire Protection, Detection and Alarm System.
- (c) The information shown on the Drawings with respect to the arrangement of the components of the fire detection and protection systems, has been arrived at after preliminary discussions between the S.O. and FSSD. This information is intended to convey to the Contractor, the minimum requirements. However, the information provided is not intended to be exhaustive. The Contractor shall be responsible for ensuring that the final design and installation work is acceptable to FSSD. The Contractor shall engage the appropriate personnel and Fire Engineer (FE) for assistance in obtaining all permits including but not limited to TFP, FSC, TOP and CSC for the Works.
- (d) The Contractor shall rectify all deficiencies identified by FSSD, or their representative Registered Inspectors (to be appointed by the Board to inspect the installed fire protection works for compliance with FSSD requirements) at any time, to allow FSSD to issue the Fire Safety Certificate (FSC) for the Works to the S.O.. This must be achieved before the occupational readiness of the buildings.
- (e) The fire detection and protection system shall be designed and coordinated in conjunction with the architectural, structural and other mechanical and electrical services in a planned and coordinated manner. The fire detection and protection systems shall include but not be limited to the following components:
 - (i) Fire alarm system;
 - (ii) Gas Suppression system;
 - (iii) Fire hose reel system;
 - (iv) Dry riser system; and
 - (v) Fire extinguishers.
- (f) The Drawings indicate conceptual design and minimum requirements. The Contractor shall provide all appurtenances and necessary provisions to provide complete, operating systems. All the systems shall be certified by the Contractor's respective Professional Engineer (PE), irrespective of whether or not all items are specifically mentioned in the Specifications and shown on the Drawings.
- (g) Where the Contractor's design required the building to have One-way Emergency Voice Communication and Two-Way Emergency Voice Communication, Fire Command Centre (FCC) and other Fire Protection services, the Contractor shall provide these in accordance with the prevailing Fire Code and is deemed to include these in the Contract Sum.

P7.6.2 Fire Alarm System

- (a) The Contractor shall provide a fully functional fire alarm system of the addressable type complete with MAP, SAP, Repeater Panels (RP), smoke and heat detectors, alarm bells, strobe lights, manual break glass call points, cabling, conduits, trunking, accessories etc. as a whole.
- (b) Fire Alarm Panels
 - (i) Communications between SAP and C4A MAP shall utilise a fire-rated fibre optic backbone provided by Contract 5A. The Contractor shall provide any networking hardware and patch-leads required to connect the fire alarm system to the backbone fibre patch panel in each facility. Existing equipment is Edwards EST system.
 - (ii) The Contractor shall liaise with Contract 4A to integrate all the SAP's and will be responsible for the integration until all the fire alarm systems are integrated.
- (c) The Contractor is to supply and install the BMS interface cabling from the MAP of this contract to the main BMS.
- (d) Smoke detectors under this Contract shall be provided in each of the Electrical Main Switchrooms and extended to the cable chambers and all relevant spaces as per Fire Code.

P7.6.3 Gas Suppression System

- (a) IG-100 inert gas suppression system to be provided in the following rooms complete with inlet nozzles, smoke detectors, oxygen gauge, gas discharge signage, siren and strobe light, evacuation signage, clean gas panel and to be linked to the fire alarm system:
 - (i) Non-Process Server and XLV Room;
 - (ii) Process Server and XLV Room;
 - (iii) UPS Room; and
 - (iv) Battery Room.
- (b) The system must be compliant with NFPA 2001 and relevant local authority having jurisdiction.
- (c) System design, including performance calculations, pipe calculations shall be submitted to S.O. for approval.
- (d) The inert gas suppression cylinders will be housed in dedicated rooms with inert gas piping into the rooms stated above as indicated in the Drawings.

- (e) All piping, fittings, brackets, clamps and accessories to be SS304.
- (f) Pressure relief dampers are to be installed, with calculations for the selection to be submitted for S.O.'s approval.
- (g) Manual gas release switch shall be installed for emergency manual activation as per NFPA 2001 guidelines.
- (h) Upon confirmation of impending gas discharge via the activation of smoke detector and the secondary confirmation smoke detector, or the activation of the manual release switch, the ACMV is to be shut down automatically via BMS prior to release of the gas.
- (i) At the entrance to each inert gas protected room installed with inert gas suppression system, a Breathing Apparatus (BA) set shall be provided. The BA shall be housed in a SS304 cabinet complete with lock and key inside a break glass. The cabinet shall be prominently mounted on the wall outside the room near to the access door. It shall be unobstructed by the opening of the door.
- (j) In the event of gas discharge within the room, the room has to be declared safe and all fire sources extinguished by authorised personnel before exhausting the Inert Gas. The exhaust fan provided will have to manually switched on to remove the IG 100 till the oxygen rises to level as indicated by oxygen gas indicator to enable the room to be declared safe by authorised person.
- (k) The Contractor shall install a supervised disconnect switch to avoid unwanted discharge of the gas during maintenance. The switch shall have the following as a minimum:
 - (i) When activated, the disconnect switch shall interrupt the releasing circuit to the suppression system.
 - (ii) The switch shall cause a supervisory signal at the releasing control unit;
 - (iii) When the switch is used, it shall annunciate at the panel to indicate an "out of service mode".
 - (iv) The disconnect switch shall be secured against unauthorised use by one of the following methods, which shall be subject to approval of the S.O.:
 - 1) Locate inside a lockable releasing control panel;
 - 2) Locate inside a lockable enclosure; and
 - 3) Require a key for activation of the switch.
 - (v) The switch shall be UL listed.
 - (vi) When the disconnect switch requires a key for activation, the access key shall not be removable while disconnected so the suppression system can be quickly returned to the operational condition in the event of a fire;

- (vii) Disconnect via software programming in lieu of a physical switch as described above is not allowed ; and
- (viii) System shall be tested and commissioned as per General Specifications, Particular Specifications and NFPA 2001, including NFPA2001 Annex C and authority having jurisdiction.

P7.6.4 Fire Hose Reel System

- (a) The Contractor shall provide a fully functional fire hose reel (HR) system complete with 30m hose reel drum, SS304 hose reel cabinet, associated pipework, valves and fittings, etc. as a whole.
- (b) Pipework material for hose reel to be of SS304 complete with SS304 fittings for elbow, tees, etc.
- (c) HR water supply shall be taken from Contract 3A HR water supply. Contractor shall liaise and co-ordinate with Contract 3A.

P7.6.5 Dry Riser System

- (a) The Contractor shall provide a fully functional dry riser system complete with dry riser stand pipe, pressure reducing valves, landing valves complete with pressure release valves, 2-way breeching inlets, stand-by hose, associated accessories, etc. as a whole.
- (b) The Contractor shall provide dry riser to each staircase core complete with 2-way breeching inlet at the 1st storey level.
- (c) Earthing shall be provided for each dry riser at the 2-way breeching inlet location.

P7.6.6 Public Address System

- (a) The Contractor shall provide a fully functional Public Address (PA) system complete with amplifiers, pre-amplifiers, IP network audio adaptor, IP network switch, box speakers, cabling, associated accessories, etc. in compliance with the One-Way Emergency Communication System described in Fire Code.
- (b) The Contractor shall terminate the system at a C5A patch panel located in the Non-Process XLV Room. The PA speakers in each building will be linked and connected to the One-Way Voice Emergency Communication System main panel located in the FCC at Industrial Liquid Module Headworks 1 (WIL11) by Contract 5A. The Contractor shall ensure full integration with the existing system (TOA, VX-3000 Series ; Japan).
- (c) The minimum sound level in the area covered shall be at least 65dBA or 10dBA above the ambient noise level, whichever is higher and shall not exceed 105dBA.

P7.6.7 Fire Extinguishers

- (a) The Contractor shall provide wall mounted fire extinguishers to all areas as per Fire Code. .
- (b) Electrical LV Switchrooms shall be provided with wall mounted carbon dioxide (CO₂) type fire extinguishers.

P7.7 PLUMBING AND SANITARY SYSTEMS

P7.7.1 General

- (a) The Contractor shall provide the following:
 - (i) Sanitary system;
 - (ii) Water plumbing system to connect to the potable water supply network provided by C1A contractor;
 - (iii) Licensed Plumber to install the works; and
 - (iv) The Plumbing System shall be complaint to SS636 and relevant PUB codes of Practice (e.g. water efficiency labelling scheme, application of water supply, SS 577 etc.).

P7.7.2 Sanitary System

- (a) The Contractor shall provide underground main drain lines, inspection chambers, above ground sanitary drainage pipe work and fittings, etc. connecting to the Plant Drainage Pump Station complete with tumbling bay / back drop, etc. as a whole. The Contractor shall coordinate with Other contractors for the connection details to the Drainage Pump Station after commencement of the Works.
- (b) All inspection chamber covers shall be of heavy duty type.
- (c) Vent pipe to be provided for 1st inspection chamber connecting to ventilating stacks and to be terminated with vent cowl at Roof Level. Vent pipes shall be provided for all wash hand basin, floor traps, urinal traps, WC for the above ground sanitary drainage system.
- (d) Rodding eye shall be provided for the purposes of maintenance in the event of blockage.
- (e) Provide uPVC pipes, minimum schedule 80, and fittings for all above ground sanitary soil waste and vent stacks and pipework serving toilets, pantry, etc as a whole.
- (f) Provide heavy duty ductile iron pipes and fittings for all underground sanitary soil waste and vent stacks and pipework serving toilets, pantry, etc as a whole.
- (g) Provide SS 316 floor traps and wastes outlet gratings to NEA approval complete with anti-mosquito devices.

P7.7.3 Water Supply Plumbing System

- (a) Provide complete cold water supply pipework and fittings and etc. connecting to the water supply network (by C1A contractor) complete with meter chamber, stop and check valves, etc. as a whole.
- (b) Isolation/stop valve is to be provided at each fitment/faucet at 2.4m level together with mini ball valve before connecting to the fittings, fitment/faucet.
- (c) Pipework for the water supply reticulation system shall be:
 - (i) Ductile Iron Cement Lined (DICL) for underground pipework.
 - 1) All underground pipes with vehicular traffic to be fully haunched.
 - (ii) SS304 for above ground pipework and fittings of all diameters.
 - (iii) Alignment of underground pipes to be marked out with stainless steel marker plates and flushed to ground.
- (d) Double check valves are to be provided to each fitment/faucet.

P7.8 TESTING AND COMMISSIONING

- (a) The Contractor shall refer to the General Specifications and in particular G2.26, and the Particular Specification P2 Preliminaries and General, for the testing and commissioning requirements of the Building Services works.

END OF SECTION